Tiling Game

Barchin na Charos barimo gukina umukino kuri grid $2N \times 2M$ unit square cells. rows zifite inomero 0 kugeza $2N-1$ uhereye hejuru ugana hasi, naho columns zifite inomero 0 kugeza $2M-1$ uhereye ibumoso ugana iburyo. Kuri $0 \leq i < 2N$ na $0 \leq j < 2M$, tugaragaza cell muri row i na columns j nka (i, j) .

Barchin ahaye Charos $N \cdot M$ **blocks**, imwe kuri imwe. Buri block ni kare ya 2×2 consisting of four 1×1 **tiles**. Barchin yasize irangi kuri buri tafari yaba umukara cyangwa umweru mu gihe yemeza ko **at least one tile is white**.

Charos agomba gushyira buri gice kuri grid **akimara kugihabwa**, atazi ibice azahabwa nyuma. Blocks ntizishobora kuzunguruka. Buri block igomba gushyirwamo yose imbere muri grid, gitwikiriye grids enye . Byongeye kandi, hejuru ibumoso bwa buri block hagomba gu coveringa a cell whose row and column coordinates are both even. Buri cell kuri grid ishobora gutwikirwa na bloki imwe gusa.

Barchin atsinda umukino iyo, nyuma yo gushyiraho bloki iyo ari yo yose, hari 2×2 square of cells that is covered by four black tiles. Mu buryo bwemewe, niba cells (a, b) , $(a+1, b)$, $(a, b+1)$, $(a+1, b+1)$ are all covered by black tiles for some $0 \leq a < 2N-1$ and $0 \leq b < 2M-1$, icyo gihe Barchin atsinda. The indices a na b ntibigomba kuba bingana.

Charos aratsinda niba ashyize $N \cdot M$ yose nta Barchin atsinze. Menya ko gushyira $N \cdot M$ blocks bizatwikira grid yose.

Inshingano yawe ni ugushyira mu bikorwa ingamba za Charos kugira ngo atsinde umukino. Bishobora kugaragara ko, hakurikijwe amabwiriza yatanzwe, Charos ashobora gushyiraho bloki kugira ngo yemeze intsinzi, hatitawe ku mabara y'bloki azabonera nyuma.

Implementation Details

Ugomba gushyira mu bikorwa inzira ebyiri:

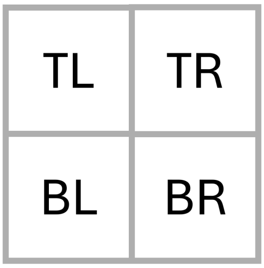
```
void init(int N, int M)
```

- N : kimwe cya kabiri cy'a rows ziri muri grid.
- M : kimwe cya kabiri cya columns ziri muri grid.

- Uburyo bwitwa rimwe gusa kuri buri kibazo cy'ikizamini, mu ntangiriro y'ishyirwa mu bikorwa rya programu yawe.

```
std::pair<int, int> receive_block(int TL, int TR, int BL, int BR)
```

- TL , TR , BL , BR : amabara y'amatafari yo hejuru ibumoso, hejuru iburyo, hepfo ibumoso, n'amatafari yo hepfo iburyo tiles of the current block, nkuko bigaragara ku ishusho iri hepfo. Agaciro kose ni 0 (umweru) cyangwa 1 (umukara).
- Ubu buryo bwitwa neza neza inshuro $N \cdot M$ kuri buri test case, nyuma yo guhamagara bwa mbere kuri `init` .



Ubu buryo bugomba gusubiza pair ya integer (i, j) , aho i ari row coordinate na j ari column coordinate ya cell aho hejuru ibumoso bwiyi bloki kagomba gushyirwa. i na j byombi bigomba kuba even, kandi agace ka 2×2 gatwikiriwe n'a bloki kagomba kudahurirana na blocki yari yashyizwemo mbere.

Niba `receive_block` igarutse ku gikoresho kitujuje ibi bisabwa, cyangwa niba nyuma yo gushyiraho agace gato k'uturemangingo tw'amadorari 2×2 gatwikiriwe n'udutafari tw'umukara, umuhanga mu by'igenamigambi azahita ahagarika gahunda yawe kandi umwanzuro w'urubanza rw'ikizamini uzaba `Output isn't correct` .

Imyitwarire ya grader ni **not adaptive**. Ibi bivuze ko urukurikirane rwa blocks Barchin aha Charos ruhinduka mbere yuko `init` ihamagarwa.

Constraints

Kuri buri gice, reka S ibe umubare wa black tiles muri ayo ma tiles ane. Ni ukuvuga, $S = TL + TR + BL + BR$.

- $1 \leq N, M \leq 100$
- $0 \leq S \leq 3$ kuri buri block.

Subtasks

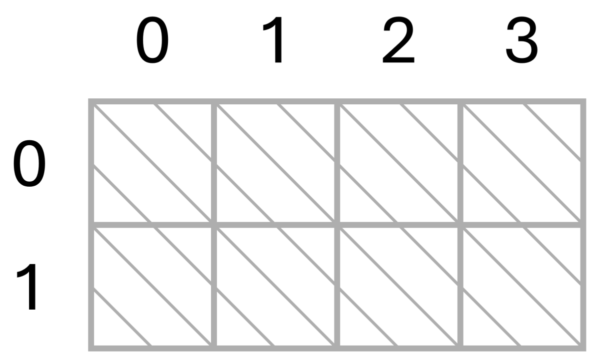
Subtask	Score	Additional Constraints
1	6	$S = 1$ for every block, and $N = 2$.
2	16	$S = 3$ for every block. $N = M$, N is even, and each of the four possible block colorings appears exactly $\frac{N^2}{4}$ times.
3	10	$S = 1$ for every block.
4	29	$S \leq 2$ for every block.
5	39	No additional constraints.

Example

Tekereza umukino ufite $N = 1$ na $M = 2$, bityo gridi ifite rows 2 na columns 4 . The grader first calls:

```
init(1, 2)
```

Mu ntangiriro, cells zose ziba zirimo ubusa. the grid imeze gutya gutya:

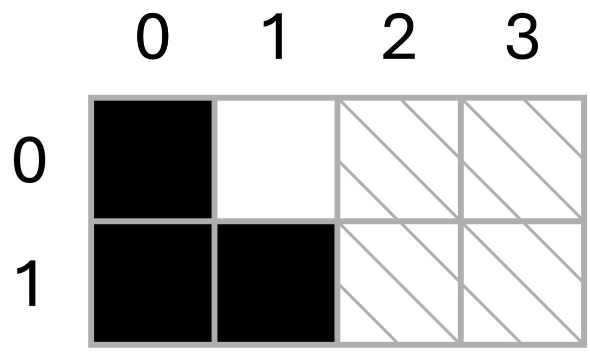


Hari $N \cdot M = 2$ blocks zo gushyiramo. Tuvuge ko Barchin atanze block with three black tile na tile imwe y'umweru hejuru iburyo mumfuruka. grader calls

```
receive_block(1, 0, 1, 1)
```

Charos ahitamo gushyira iyi bloki ibumoso bwa grid by returning $(0, 0)$.

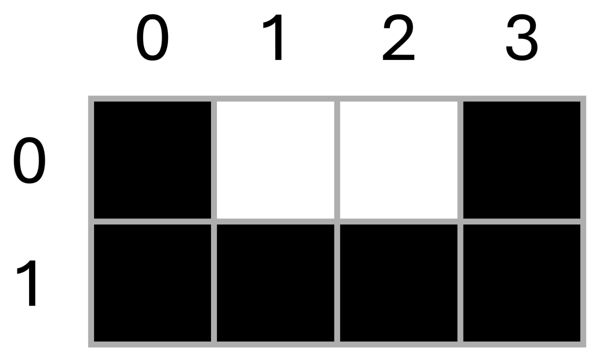
The grid now looks like this:



Barchin then gives another block with three black tiles and one white tile at the top-left corner:

```
receive_block(0, 1, 1, 1)
```

cell imwe rukumbi isigaye ifite even row na even column gukora nk'imfuruka yo hejuru ibumoso ya block ya 2×2 ni $(0, 2)$, bityo Charos returns $(0, 2)$. The grid ends up like this:



Nta 2×2 gitwikiriwe na black tiles, bityo Charos ya shizeho neza bloki zose nta Barchin na rimwe atsinze. Charos ni we utsinda umukino.

Sample Grader

Input format:

```
N M
TL[0] TR[0] BL[0] BR[0]
TL[1] TR[1] BL[1] BR[1]
...
TL[NM-1] TR[NM-1] BL[NM-1] BR[NM-1]
```

Output format:

```
R[0] C[0]
R[1] C[1]
...
R[NM-1] C[NM-1]
```

Hano, $R[k]$ na $C[k]$ ni pair za integer returned by k -th call kuri `receive_block` .